How Effective Has Industrial Policy Been in the United States? A Review*

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Abstract

The United States returned openly to industrial policy in 2022, with semiconductor subsidies under the CHIPS and Science Act and clean-energy credits under the Inflation Reduction Act; governments across the world, including in many developing countries, are weighing similar instruments. We survey the empirical evidence on US industrial policy, drawing on a companion measurement paper (Erten et al., 2026), with the aim of identifying broadly applicable lessons. The literature has found high social returns to nondefense public R&D, substantial investment responses to tax incentives, smaller and contested spillovers from defense R&D, and local gains but uncertain aggregate effects from place-based policies. We also identify an *evaluation gap*: many substantial components of US industrial policy have not received extensive empirical scrutiny. The policy instruments with causal evidence that we would consider strong account for only about a third of the measured expenditure portfolio; the rest has been studied only in part, or with conflicting results. This includes federal credit programs (a class of instruments that appears to have been very effective in East Asia), which have been studied as small-business and export finance but not for their effects on innovation, productivity, or the structure of production. We also review evidence from other industrialized and developing countries and consider the relative merits of different instruments.

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1 Introduction

The CHIPS and Science Act of August 2022 put $52.7 billion behind the domestic semiconductor industry, $39 billion of it manufacturing incentives; the Inflation Reduction Act (IRA), enacted a week later, added open-ended clean-energy tax credits with much larger projected potential costs. The two laws were widely viewed as marking the "return" of industrial policy after decades of official disrepute. The United States is hardly alone. The Global Trade Alert database records thousands of new subsidy and local-content interventions worldwide, with advanced economies rather than developing countries accounting for most of the increase (Evenett and Fritz, 2022; Juhász et al., 2025).

Research has shifted along with the policy. There is a longstanding literature justifying industrial policies in terms of market failures, especially those associated with spillovers from learning and innovation (Arrow, 1962a,b; Stiglitz, 1975, 1987; Dasgupta and Stiglitz, 1988). Greenwald and Stiglitz (2006) and Stiglitz and Greenwald (2014) in particular emphasize learning externalities: sectors with steep learning curves (semiconductors, aerospace, pharmaceuticals) often generate knowledge benefits to other sectors, and these benefits are under-provided by competitive markets.[1] Griliches (1992) documented R&D spillovers substantial enough to justify public subsidy. A new literature, synthesized by Juhász, Lane, and Rodrik (2024), has moved the debate from whether governments should pursue industrial policy at all to the conditions under which particular instruments work. The World Bank, for decades skeptical of industrial policy, has now issued a Policy Research Report on recommended approaches to industrial policy for development (Fernandes and Reed, 2026).

In a companion paper (Erten et al., 2026), we argue that the "return" framing is misleading and that in fact the United States practiced industrial policy throughout the postwar period. Combining on-budget appropriations and off-budget tax expenditures from 1962 to 2024, federal backing for private investment and innovation averaged 0.6–1.0 percent of GDP by decade under a broad definition, and 1.0–1.9 percent with defense R&D added. What moved over the period was less the scale of intervention than its visibility. Direct appropriations dominated the Cold War decades, then ceded ground to tax-code provisions that drew less oversight.

[1] These arguments modify in an important way the longstanding infant-industry argument for protection and industrial policy: if that were the only problem, then companies could borrow on the prospect of future profits, and the intertemporal problem posed by learning would have been easily solved. But as Dasgupta and Stiglitz (1988) also point out, capital markets are far from perfect.

In this paper, as a complement to the companion paper, we take stock of what is known about the effectiveness of US industrial policy. We consider the effects of industrial policy on innovation, productivity, investment, employment, and regional development. The review is organized by policy instrument, following the breakdown of actual US spending used in the companion paper. For each category of instruments, we focus on studies that identify causal effects through quasi-experimental designs or estimated structural models. Where they seem especially relevant, we also draw on theoretical, historical, and case-study research, especially where causal studies are absent. This is not a systematic review or meta-analysis; we emphasize studies which in our judgment are the most credible, instrument by instrument. We also evaluate evidence from other industrialized and developing countries and discuss the implications especially for developing countries seeking to design policies for the clean-energy transition.

A key issue is how to define "effectiveness" of an industrial policy. We break the question into three steps. First, did the policy change *any* outcomes? Second, did it change the outcomes with which we are concerned (innovation, productivity, investment, employment, and/or regional development)? Third, did the policy improve overall welfare? Closely related to this third question is the question of whether the policy helped to remedy a market failure. The literature has made progress on the first two questions but, in our view, has found the third harder. Over the course of the review, we aim to highlight areas where we think additional research would be particularly valuable.

A key finding of our review is that there is an *evaluation gap*: many components of US industrial policy have not received extensive empirical scrutiny. The quasi-experimental literature on industrial policy has concentrated on R&D tax credits, direct R&D appropriations, and place-based programs with clean eligibility thresholds. But these instruments constitute only a portion of the measured total. Much of industrial policy has been evaluated on some margins but not others, and some of what we argue should be considered industrial policy has not been rigorously evaluated at all. The most credible evidence concerns the public financing of research and tax-based investment incentives.

Our survey complements recent syntheses of the industrial-policy literature (Lane, 2020; Juhász, Lane, and Rodrik, 2024; Criscuolo et al., 2022). First, by anchoring the review to a comprehensive measurement of US industrial policy, we can ask how much of the spending the evidence covers. Coverage matters because findings for one instrument need not carry over

to another. What we know about R&D credits tells us little about loan guarantees. Second, the US experience bears on debates well beyond its borders. Often taken to be an exemplar of market-led development, the US has nevertheless continuously practiced substantial industrial policy, and the evidence that targeted support to sectors with strong learning spillovers can have large and persistent effects, as in the East Asian experience (Lane, 2025; Amsden, 1989; Stiglitz, 1996; Wade, 1990), is relevant to the choices now facing developing economies in the clean-energy transition. Tax expenditures play the role in the United States that directed credit and special-economic-zone incentives play elsewhere, as channels of support that draw less scrutiny than direct spending. The stacked subsidies of the CHIPS and IRA Acts have their counterpart in the overlapping incentive schemes common in developing economies. We focus on fiscal instruments and do not evaluate trade policies as industrial policy; trade policy in that role, including the post-2018 tariff revival, is treated in the companion paper (Erten et al., 2026), and the evidence on trade-based instruments is surveyed in Juhász, Lane, and Rodrik (2024).

The remainder of the paper proceeds as follows. Section 2 reviews the empirical evidence by instrument: sectoral incidence and descriptive patterns, government R&D spending, R&D tax credits, investment incentives, place-based policies, and the recent CHIPS Act and IRA. Section 3 quantifies the evaluation gap and asks why it exists and how much it matters. Section 4 reviews evidence from other countries, industrialized and developing. Section 5 weighs the relative merits of different instruments. Section 6 concludes.

2 Empirical Evidence on the Effectiveness of Industrial Policy

This section reviews the evidence instrument by instrument, following the categories in which the companion paper measures industrial-policy spending (Erten et al., 2026); we first summarize its definitions. We then describe how support is distributed across sectors, and why simple comparisons of support and sectoral growth say little about the effects of policy (Section 2.1). The subsections that follow take up government R&D spending; tax incentives for research and for investment, which with R&D spending are the most thoroughly studied instruments; place-based policies; and the early evidence on the CHIPS and Science Act and the Inflation Reduction Act. The federal credit programs, which have been studied mainly as small-business and export finance, are discussed in Section 3.

The companion paper measures US industrial policy under four nested definitions, which we use throughout. The *narrow* measure is civilian federal R&D outlays plus the research and advanced-manufacturing tax credits (0.46 percent of GDP in FY2024). The *broad* measure adds the targeted energy and production tax provisions, among them the deduction for foreign-derived intangible income (FDII, §250), and outlays other than R&D in the energy and the science and space budget functions (0.68 percent). Adding Department of Defense outlays for research, development, test and evaluation (RDT&E) gives *broad plus defense* (1.16 percent), and adding the face value of federal credit programs gives what we call the *comprehensive* portfolio (1.31 percent, or about $379 billion). These are the headline definitions; general investment incentives such as accelerated depreciation are tracked separately, outside them.[2] The definitions differ less in the instruments they include than in our ability to identify in the data what might appropriately be labeled industrial policy. Some defense expenditures are industrial policies, others are not, and the same is true of tariffs. As to instruments, there are only a few, namely tax provisions, subsidies and other direct outlays, credit, procurement, and trade measures, and each of the definitions combines several of them. Table 1 gives the breakdown by instrument.[3]

2.1 Sectoral Incidence and Descriptive Patterns

Industrial policy operates through a wide range of tax expenditures and direct subsidies, but its incidence is uneven across sectors. Combining Treasury tax expenditure estimates with industry value added from the Bureau of Economic Analysis (BEA) and mapping Internal Revenue Code (IRC) provisions[4] to NAICS sectors, we find support concentrated in manufacturing and in energy, though the available data measure it differently in each. In manufacturing, bonus deprecia-

[2]The Investment Tax Credit was for a time the largest instrument in the tax code affecting capital formation. As a general subsidy, available for any qualifying investment, it falls outside the companion paper's headline broad definition. It was repealed before modern quasi-experimental methods matured, although Cummins, Hassett, and Hubbard (1994) exploit tax reforms spanning ITC episodes, and Goolsbee (1998b) shows that part of the credit was captured in higher prices of capital goods.

[3]Other estimates differ mainly in what they count. DiPippo, Mazzocco, and Kennedy (2022) put US industrial-policy spending at 0.39 percent of GDP in 2019. They count only government support for R&D performed by business (0.15 percent of GDP), where we count all federal R&D outlays, and they value credit by the implied interest subsidy (less than 0.01 percent) and not, as we do, at face value; these two differences account for most of the gap. In the OECD's Quantifying Industrial Strategies figures for eleven OECD economies, which leave out defense, grants and tax expenditures averaged 1.53 percent of GDP in 2022; adding financial instruments at face value, as we add credit to our US measures, gives 3.65 percent, against 0.83 percent for the United States in FY2024 on a comparable basis (the broad measure plus credit at face value; OECD, 2025; Erten et al., 2026).

[4]We refer to tax provisions by their section of the Internal Revenue Code. §41 is the research credit. §168(k) is "bonus" depreciation, which lets a firm deduct much of the cost of equipment in the year it is bought, and §179 is a similar allowance for smaller firms. §45X is a credit per unit of clean-energy components made in the United States, and §48D a credit for investment in semiconductor plants.

tion (§168(k)) deductions averaged about $193 billion a year over 2018–2022, roughly 6.7 percent of the sector's 2024 value added. Since a deduction is worth only the tax it saves, and accelerated depreciation mainly changes when that tax is paid, their revenue cost is well under 1 percent of value added. In energy and utilities, the clean energy credits expanded under the IRA (§45, §48, §45Q, §45V, §48C) have a revenue cost of $21.3 billion in FY2024, roughly 4.7 percent of sector value added. Measured this way, targeted support is most intense in energy.[5]

Bonus depreciation under §168(k) is sector-neutral in rule but not effect.[6] The Joint Committee on Taxation released industry-level usage data covering tax years 2018–2022 (Joint Committee on Taxation, 2025). Manufacturing accounted for 20.3 percent of all bonus depreciation claimed over the five years; real estate and rental/leasing 15.2 percent; wholesale and retail trade 14.5 percent; and information 10.8 percent. Information rose from approximately $82 billion of bonus depreciation in 2018 to $136 billion in 2022, as cloud data centers and telecommunications networks expanded.[7] The data end before the surge in data-center investment for artificial intelligence, and the sector's share is probably larger now. These instruments thus tilt toward capital-intensive manufacturing and technology infrastructure, the sectors that explicit industrial-policy targeting tends to favor. Real estate's prominence among the claimants recalls a point developed in the companion paper. Mortgage guarantees, the government-sponsored enterprises, and the tax preferences for owner-occupied housing tilt capital toward real estate and finance rather than industry, in effect an *anti-industrial* policy, distorting patterns of investment away from productive investments with high learning and spillovers to other sectors, with potentially adverse effects on growth;[8] a portion of the measured industrial-policy portfolio works simply to offset

[5]These ratios are approximate. The credits are valued at their FY2024 revenue cost in the companion paper's data ($17.4 billion of the $21.3 billion is the §48 credit), and bonus depreciation at the deductions claimed in tax years 2018–2022 in JCT data. Value added is for 2024, from the BEA GDP-by-Industry accounts (manufacturing $2,881 billion; utilities $454 billion). The energy ratio can err in either direction: it divides by value added in the whole utilities sector although the credits go to clean energy, and some of the credits are claimed outside the sector (§48 for generation owned by other businesses, §48C by manufacturers).

[6]Accelerated depreciation does affect the structure of production, since it affects investments with different patterns of depreciation differently, but that was not its intent, and it is questionable accordingly whether one should consider it as industrial policy.

[7]In tax year 2022, oil and gas accounted for only 2.6 percent, defense contractors 1.5 percent, and pharmaceuticals 0.8 percent (Joint Committee on Taxation, 2025, Table 5). Regulated electric utilities account for little because, since the 2017 tax act, property used in rate-regulated utility businesses has not been eligible for bonus depreciation. Bonus depreciation phased down after 2022 until §70301 of the 2025 budget reconciliation act restored it, permanently and at 100 percent, for property acquired after January 19, 2025.

[8]For a theoretical treatment, see Hirano and Stiglitz (2025). Credit allocation is an important determinant of this sectoral pattern, with (non-causal) evidence of significant effects on growth (Verner, 2022; Müller and Verner, 2024). There is a large literature on the tax preferences for real estate. To our knowledge, little of it is based on causal studies in the modern sense, but its findings are what one would expect from standard price elasticities. The preferences favor housing over business capital, and leave the housing stock larger, and the stock of other capital smaller, than it would otherwise be (Mills, 1987; Gervais, 2002). The most direct causal evidence concerns credit: Chakraborty, Goldstein, and MacKinlay

these distortions (Erten et al., 2026).

Comparing support with sectoral growth says little about the effects of policy. Such comparisons have no counterfactual, support may follow growth rather than cause it, and measured real growth in computers and electronics, among the most supported sectors, largely reflects hedonic quality adjustment of computer and semiconductor prices (Houseman et al., 2011). There is a small but growing literature that uses quasi-experimental econometric techniques to assess the causal relationship between government support and sectoral growth. These attempts have exploited eligibility thresholds, discretionary award processes with close winners and losers (Howell, 2017), and narrative or legislative shocks to sectoral support (Fieldhouse and Mertens, 2024; Lane, 2025).

2.2 Evidence on the Effectiveness of Government R&D Spending

The social returns to R&D exceed private returns because firms cannot fully appropriate the benefits of the knowledge they generate (Stiglitz, 1975; Griliches, 1979; Jones and Williams, 1998; Bloom, Schankerman, and Van Reenen, 2013; Jones, 2022). Government R&D spending, particularly when directed toward basic and general-purpose research, can generate large welfare gains by correcting the underinvestment.

Whether federal R&D spending raises private-sector productivity in practice depends on the composition of the spending. Fieldhouse and Mertens (2024) construct a narrative dataset of 257 significant policy changes in the R&D appropriations of five agencies (DoD, DoE, NASA, NIH, NSF) from 1947 to 2019, covering roughly 90 percent of total federal R&D in the United States. They estimate gross social returns of 140–210 percent for *nondefense* public R&D.[9] Fieldhouse and Mertens find no measurable productivity spillovers from defense R&D within 15 years, a result they attribute in part to defense spending being tilted toward development rather than research, and in part to the slower and more limited dissemination of military technology. Fieldhouse and Mertens cover only *direct federal R&D appropriations*, a category amounting to about

(2018) show that US banks exposed to housing booms shifted lending from commercial loans to mortgages, and that firms borrowing from those banks invested less.

[9]Gross social returns are aggregate benefits, including spillovers to other firms and industries, beyond the private returns captured by the performing firms. Fieldhouse and Mertens (2026) apply these estimates to CHIPS Act projections, and the framework the Congressional Budget Office has developed for estimating the economic effects of federally funded R&D relies heavily on them (Campbell et al., 2026).

0.8 percent of GDP in FY2024.[10] The tax expenditures and credit programs that make up the rest of US industrial policy lie outside their data.

Kantor and Whalley (2025) focus still more narrowly, studying NASA contractor spending (NASA's budget exceeded 0.7 percent of GDP in the mid-1960s) using declassified CIA Soviet technology assessments as an instrument. They find a fiscal multiplier of only about 0.3, large effects on value added and employment in directly treated county-industries, and no gains in total factor productivity (TFP) or cross-industry spillovers.[11]

Other studies of US defense spending find larger effects on innovation, though not all of them are spillovers in the strict sense. Draca (2013) documents, exploiting firms' differential exposure to the Reagan-era procurement build-up, that it raised patenting and R&D among US defense contractors, and Howell et al. (2025) show that reforms opening the Air Force's Small Business Innovation Research (SBIR) competitions to firm-initiated "open" topics generated commercial innovation and military technology adoption that narrowly specified topics did not. Evidence from a broader range of advanced countries also points to positive spillovers from defense R&D. Moretti, Steinwender, and Van Reenen (2025), using industry-level variation across OECD countries and firm-level data from France, find that public defense R&D can crowd *in* privately financed R&D and raise industry productivity.

Much of this defense evidence is, in effect, evidence on procurement, an instrument the companion paper's fiscal portfolio captures only in part. Defense R&D contracts are in it, but purchases of finished goods are not, since a purchase is not a subsidy and enters industrial-policy accounting only through any premium paid over market prices, which is rarely observed. The causal evidence for procurement extends beyond defense (for a survey, see Chiappinelli, Giuffrida, and Spagnolo, 2025). Slavtchev and Wiederhold (2016) find that shifting the technological content of federal procurement toward high-tech goods raises private R&D across US states, and Hebous and Zimmermann (2021) show that federal procurement shocks raise investment by financially constrained firms. In historical work, Clemens and Rogers (2026) show that Civil War procurement policies shaped the direction of medical innovation in prosthetics.

Government grants have well-documented effects on innovation outcomes such as patent-

[10] Civilian R&D outlays of 0.32 percent and Department of Defense RDT&E outlays of 0.48 percent. Together they are 69 percent of the broad-plus-defense measure (1.16 percent of GDP) and about three-fifths of the comprehensive measure, which adds the federal credit programs (1.31 percent).

[11] Small multipliers may also reflect that unemployment was not high at the time; multipliers depend heavily on the economy's excess capacity.

ing, as noted above for defense (Howell et al., 2025). Using quasi-experimental variation in National Institutes of Health (NIH) funding, Azoulay et al. (2019) find that public research funding to academic scientists generates significant downstream patenting by private firms in pharmaceuticals and biotechnology, evidence of knowledge spillovers from publicly funded science.[12] In historical work, Gross and Sampat (2025) show that targeted federal investments in biomedical research during World War II produced lasting increases in US innovation and research capacity through the 1950s and 1960s.[13] For energy innovation programs targeting small firms, Howell (2017) finds that Department of Energy SBIR awards increase patenting and approximately double the probability that recipient firms secure venture capital, relative to firms that marginally miss the award; Myers and Lanahan (2022) estimate that for every patent produced by grant recipients, three more are produced by others who benefit from spillovers.

2.3 Evidence on the Effectiveness of Tax Incentives for Research and Investment

The R&D tax credit (§41), a credit for qualified research spending above a base tied to the firm's own past research, has been a central instrument of US industrial policy since its creation in 1981, as documented in the companion paper (Erten et al., 2026). In real FY2024 dollars, the credit grew from about $1 billion in 1982 to about $20 billion in 2023, while annual utility patent grants rose from roughly 58,000 to 314,000; the levels correlation between the two series exceeds 0.9, but year-to-year changes are only weakly correlated ($r \approx 0.2$).[14]

Earlier studies of R&D tax credits used cross-country panel data (Bloom, Griffith, and Van Reenen, 2002),[15] documenting positive associations between changes in tax rules and changes in R&D spending.[16] Using state-level variation from 1981 to 2004 in R&D tax incentives, Wilson (2009) finds that these incentives are effective in increasing in-state R&D, but that nearly all of the increase comes from geographic reallocation away from less generous states, with no

[12] Li, Azoulay, and Sampat (2017) document that only about 10 percent of NIH grants generate a patent directly, but roughly 30 percent produce research later cited by patents: the impacts diffuse beyond the original grantees.

[13] Gross and Sampat (2023) show that the wartime R&D expansion administered by the Office of Scientific Research and Development created persistent regional innovation hubs and shaped postwar US technological development.

[14] Utility patents cover new and useful inventions or processes, as distinct from design and plant patents. Real §41 tax expenditure (GDP-deflator adjusted) and utility patent grants (PatentsView), computed from the replication dataset of Erten et al. (2026): levels $r = 0.92$ and year-over-year changes $r = 0.21$ over 1982–2023.

[15] For surveys of this literature, see Hall and Van Reenen (2000) and Becker (2015).

[16] Reviewing the macro and micro evidence, Bloom, Van Reenen, and Williams (2019) conclude that a 10 percent decline in the tax cost of R&D is associated with at least a 10 percent increase in R&D over time, a long-run elasticity of roughly one or higher.

change in aggregate R&D spending.[17] More recent studies exploit firm-level quasi-experimental variation created when tax policy changes affect some firms more than others, for instance bunching and regression discontinuity designs around thresholds for favorable tax treatment (Chen et al., 2021; Dechezleprêtre et al., 2023). Some firms respond to such incentives by relabeling existing expenditures as "research and development,"[18] but there is nevertheless broad evidence that R&D tax credits have positive effects on patents, productivity, and employment.

For the United States, Rao (2016) uses tax-rule changes in 1981–1991 to identify changes in each firm's cost of R&D. A 10 percent drop in that cost raises the average firm's R&D intensity (R&D relative to sales) by 20 percent in the short run, and by more in the long run, though small and young firms partly reverse their initial increases. In 25 states over 1990–2010, Fazio, Guzman, and Stern (2020) find that introducing a state R&D tax credit raises new business formation by about 7 percent at first and by roughly 20 percent over ten years, in both quantity and quality-adjusted terms. Lucking (2019) finds that states offering R&D tax credits experience higher employment growth, reflecting real expansion at R&D-performing firms rather than jobs shifting across states. Wilson's reallocation result and Lucking's are not in conflict, since they concern different margins: R&D spending, which moves easily across state lines, versus employment at R&D-performing firms, which is far less mobile.

The other large group of tax incentives is for investment. Investment tax credits, accelerated depreciation, and related deductions, whose history the companion paper traces, were not designed to affect the structure of production and lie outside our headline definitions, but their scale is large (Erten et al., 2026). These policies can raise investment in equipment and structures (Cummins, Hassett, and Hubbard, 1994; Zwick and Mahon, 2017; Maffini, Xing, and Devereux, 2019).[19] Ohrn (2019) exploits state-level variation in the adoption of accelerated depreciation in a difference-in-differences framework and finds that it increases capital investment in US manufacturing. Ohrn (2018) finds comparable responses to the Domestic Production Activities Deduction (§199, in effect 2005–2017), a manufacturing-directed provision whose revenue cost peaked at roughly $20 billion in FY2016, about 0.11 percent of GDP. A one percentage point reduction in

[17] Bloom and Griffith (2001) document a similar reallocation across countries: R&D in one country responds to the price of R&D in competitor countries.

[18] Chen et al. (2021) estimate that relabeled expenses account for approximately 24 percent of reported R&D investment in response to a change in corporate tax rules in China.

[19] For earlier studies on the investment effects of tax policy, see Hall and Jorgenson (1967), Summers (1981), Hayashi (1982), and Chirinko, Fazzari, and Meyer (1999).

effective tax rates raised investment by 4.7 percent of installed capital. The 2017 tax act, which combined immediate expensing of equipment with a lower corporate rate and new international provisions, is the most recent large test. Chodorow-Reich et al. (forthcoming) find that C corporations facing the mean tax change raised domestic investment by 18 percent relative to firms with no change. In the general-equilibrium model they calibrate to these estimates, domestic corporate capital is about 6 percent higher in the long run and wages about 0.9 percent higher, while the added growth recovers only 7 percent of the revenue lost.

The labor-market effects are less clear. Tuzel and Zhang (2021), studying §179 expensing with establishment-level occupational data, find that investment incentives raise equipment investment among eligible firms but leave total employment roughly unchanged; instead they shift its composition, increasing skilled employment in the short run while reducing routine-task employment over subsequent years. However, Curtis et al. (2022), using Census microdata on manufacturing establishments, find that lower capital costs raise both investment and overall employment, concentrated among production workers, through lowering production costs, as firms expand output and demand for all inputs rises. Evidence on the 2017 cut in the corporate tax rate, a general tax cut rather than an investment incentive, shows that the gains to workers can be narrow. Kennedy et al. (2026) compare C corporations, whose top rate fell from 35 to 21 percent, with S corporations of similar size in the same industry, and obtain investment elasticities similar to those of Chodorow-Reich et al. (forthcoming). Relative to the S corporations, the C corporations raised their capital stock by 4.6 percent, payroll by 2.0 percent, and employment by 1.3 percent, but earnings rose only for the best-paid tenth of their workers, and most for executives. In the short run, the authors estimate, 87 percent of the gains in private income went to the top 10 percent of the income distribution.

These results should not be a surprise: these tax provisions lower the cost of capital relative to labor, so we should expect substitution against labor. If skilled labor is more complementary with capital, then we would expect an increase in skilled labor input and a decrease in unskilled labor. Confounding these microeconomic effects are macroeconomic effects, especially when the economy is significantly short of full employment; then the tax benefits stimulate the economy, unless offset by tax increases elsewhere.

2.4 Evidence on the Effectiveness of Place-based Policies

Place-based policies in the United States, such as Empowerment Zones, enterprise zones, and geographically targeted investment subsidies, have been widely used to stimulate economic activity in the targeted locale, often in distressed regions. Aggregate efficiency and welfare effects depend on spillovers, migration responses, and whether new activity represents net national gains or spatial reallocation. Much of the causal literature has, however, focused on the narrower question of whether the policies succeeded in promoting growth in the targeted area. Exploiting rejected applicants as a comparison group, Busso, Gregory, and Kline (2013) find that federal Empowerment Zone designation increased employment and wages in treated areas, with little evidence that the gains were driven solely by changes in population composition. Kline and Moretti (2014) study the Tennessee Valley Authority, one of the largest place-based development programs in US history; they find that large-scale public investment in distressed regions generated persistent increases in manufacturing employment and long-run improvements in regional outcomes. The broader literature reviewed by Neumark and Simpson (2015) is more guarded, particularly on state enterprise zones, concluding that the magnitude and persistence of effects vary widely across programs and settings. More recent evidence on Opportunity Zones suggests that while place-based tax incentives can generate large investment flows, much of the investment may be inframarginal (activity that would have occurred anyway), with limited evidence of improvements in resident wellbeing (Corinth and Feldman, 2024).

Using quasi-experimental variation from the California Competes Tax Credit, Freedman, Khanna, and Neumark (2023) find that awards raise employment in targeted tracts, with gains extending well beyond the directly incentivized jobs, at modest direct cost per pledged job. The theoretical case for intervention rests on agglomeration externalities. Greenstone, Hornbeck, and Moretti (2010), comparing counties that won large plant openings to runner-up counties in the same site-selection process, document total factor productivity gains of roughly 12 percent among incumbent plants in winning counties five years after the opening, larger for plants that share labor and technology pools with the new plant. Competition for these plants across counties hands much of the gain to the firms. Using data on the subsidy deals that states offered to attract large plants, Slattery (2025) finds that competition among states raises welfare by less than 5 percent relative to a ban on such subsidies, because states compete away the surplus

and transfer the rents to the firms. These designs identify average effects and who captures the gains; they offer less guidance on what makes a program succeed. As Hanson, Rodrik, and Sandhu (2025) emphasize, local organizational capacity conditions effectiveness, since many modern place-based policies channel resources through decentralized "policy supply chains" of intermediary organizations.

2.5 Evidence on the Effectiveness of the CHIPS and Science Act

The objective of the CHIPS and Science Act was to promote innovation, especially in critical areas like semiconductors, and to reduce US dependence on imported chips. It is too early to evaluate the Act against these aims; the evidence so far consists of projections and of early effects on employment. Fieldhouse and Mertens (2026) estimate that the Act's authorized R&D funding could increase US productivity by approximately 0.2–0.4 percent over the medium term, with annual output gains exceeding $40 billion at peak impact. These are projections that apply their causal appropriations-shock estimates to the Act's funding levels, and they align with the broader evidence that public R&D generates large social returns through spillovers (Bloom, Schankerman, and Van Reenen, 2013; Jones, 2022; Fieldhouse and Mertens, 2024). The semiconductor sector is likely to exhibit such spillovers because innovation is cumulative and highly interdependent across firms and stages of the supply chain. International evidence from Taiwan, Korea, and China suggests that subsidies, research collaborations, and industrial policies played an important role in the development of semiconductor industries (Goldberg et al., 2024). At the same time, international spillovers imply that the benefits of national semiconductor subsidies may partly accrue to foreign firms and consumers.

Using a dynamic oligopoly model calibrated to foundry-level data, Miao (2024) estimates that the 25 percent investment subsidy authorized under the CHIPS Act is optimal if the risk of trade disruption is put at 35 percent, in which case it raises US welfare by about 6 percent, as subsidies increase domestic capacity, strengthen supply-chain resilience, and raise firms' incentives to invest in advanced technologies. That gain is the model's valuation of the insurance that domestic capacity provides against supply disruption, not a forecast of GDP, and it rises and falls with the assumed probability of disruption. The result echoes the older theoretical case for intervention in industries with technological externalities, dynamic learning, and strategic competition (Brander

and Spencer, 1985; Eaton and Grossman, 1986; Aghion et al., 2001; Greenwald and Stiglitz, 2006; Stiglitz and Greenwald, 2014; Juhász, Lane, and Rodrik, 2024).

While the CHIPS and Science Act was not motivated by employment creation, it is already generating measurable employment effects. In companion work using county-level difference-in-differences designs, Erten, Stiglitz, and Verhoogen (2025) find that semiconductor-producing counties experienced significant short-run employment gains following Senate passage of the United States Innovation and Competition Act (USICA),[20] with increases of roughly 110–180 semiconductor jobs per county, representing growth of approximately 12 percent relative to pre-policy levels, plus spillovers into construction and related sectors. Summed across counties, their estimates imply direct gains of roughly 15,000–16,000 semiconductor jobs and 15,000–30,000 jobs in related sectors; because the design compares treated with untreated counties, these totals are gross gains in treated areas and cannot rule out offsetting losses elsewhere. These are anticipation effects: the identifying event precedes enactment, consistent with evidence that economic activity responds to fiscal news before spending occurs (Ramey, 2011; Nakamura and Steinsson, 2014).

2.6 Evidence on the Effectiveness of the Inflation Reduction Act (IRA)

The Inflation Reduction Act is a historically large industrial policy whose effects operate through several channels: accelerated deployment of clean electricity, supply-side expansion of energy capital, changes in electricity market prices, and the reshoring of clean-energy manufacturing. Because the policy is recent, much of the early assessment is structural, but it builds on a large empirical literature on renewable subsidies. We know of no causal evidence yet on its manufacturing credits (§45X and §48C).

Bistline, Mehrotra, and Wolfram (2023) model the IRA's tax credits for clean electricity, electric vehicles, and manufacturing and estimate that the climate provisions could reduce US greenhouse gas emissions to 32–42 percent below 2005 levels by 2030, with total fiscal costs of $900 billion to $1.2 trillion over the decade. Their analysis shows that production tax credits, paid per unit of output, lower the effective marginal cost of renewable generation and can induce negative wholesale electricity prices, while investment credits subsidize upfront capital costs, working instead through entry and capacity; the result is a significant expansion of renewable capacity and

[20] USICA was never enacted as such; it was later folded, in pared-down form, into the CHIPS and Science Act.

lower retail electricity prices, transferring surplus from taxpayers to electricity consumers and renewable producers. They estimate an average abatement cost of roughly $45–61 per ton of CO_2 for the power-sector credits, well below recent estimates of the social cost of carbon, such as the $185 per ton estimated by Rennert et al. (2022). Related work reaches similar conclusions (Jenkins et al., 2022; Zhao et al., 2022; Bistline et al., 2023), finding that uncapped credits substantially accelerate renewable deployment, but fiscal costs and emissions reductions depend critically on financing conditions, technology costs, and supply constraints.

There is by now a substantial body of evidence, much of it causal, on the effects of green industrial policies. Gerarden, Reguant, and Xu (2026) synthesize the literature on renewable subsidies, concluding that policy, including production and investment tax credits similar to those expanded under the IRA, deserves part of the credit for falling costs and faster deployment, consistent with earlier studies such as Hitaj (2013) and Gerarden (2023), and with work documenting learning-by-doing and policy-induced cost declines (Bollinger and Gillingham, 2026; Bradt, 2024). The form of the subsidy also matters: wind farms that took an upfront investment grant in place of the production credit generated 10 to 12 percent less electricity than they would have under the production credit (Aldy, Gerarden, and Sweeney, 2023). Hahn et al. (2026) show that renewable subsidies deliver high social returns relative to other climate policies, reflecting both environmental benefits and technological spillovers. Subsidies can also reshape global production. China's large-scale solar subsidies contributed to global cost declines and supply-chain concentration (Bañares-Sánchez et al., 2026).

The first evidence on the IRA's labor-market effects concerns the clean-electricity tax credits, which account for roughly half of the law's tax-credit spending in the estimates of Bistline, Mehrotra, and Wolfram (2023). In a synthetic difference-in-differences design comparing counties with high propensity for solar and battery development (measured from solar irradiance and pre-existing transmission capacity) to others, Shimberg (2026) finds that by mid-2025 employment in directly related construction industries had risen by nearly 100 jobs per county in the most exposed group (the 786 counties with above-median irradiance and transmission capacity), an 18 percent increase over the pre-IRA mean, with spillovers of roughly 90 further jobs in related construction. Aggregated across the counties with above-median irradiance or transmission capacity, the combined gains amount to about 190,000 jobs, or 2.2 jobs per megawatt of new capacity. The estimates are local gains rather than net national job creation. Shimberg shows that

because transmission capacity is concentrated in populous counties whose residents already favored climate policy, the employment gains accrued disproportionately to existing supporters, limiting the credits' capacity to build new political constituencies.

A substantial amount of IRA funding went to electric vehicles (EVs). Using variation in eligibility rules, Allcott et al. (2024) find that the credits shifted buyers sharply toward leasing and that consumers captured a meaningful share of them, although 72 to 84 percent of recipients would have bought an EV anyway. Projections suggest that repealing the credits would nonetheless cut EV sales substantially,[21] which is likely to weaken incentives for investment in domestic EV production and supply chains, although these demand-side estimates do not directly identify effects on production investment or supply-chain capacity.

3 The Evaluation Gap

By the *evaluation gap* we mean the part of US industrial policy whose effects have not been credibly evaluated. Section 3.1 measures it, by setting FY2024 spending on each instrument against the strength of the causal evidence reviewed in Section 2. Section 3.2 asks why the gap exists and how much it matters.

3.1 Measuring the Gap

Table 1 maps the FY2024 composition of the comprehensive industrial-policy portfolio measured in the companion paper to the strength of the causal evidence for each instrument category. The table addresses two distinct questions. The first is the state of the causal evidence, that is, whether an instrument has been credibly evaluated, whether the credible studies agree, and whether they measure the outcomes that motivate treating the instrument as industrial policy, such as innovation, productivity, or industrial capacity. The "causal evidence" column grades these, and a well-identified null counts as strong evidence. The second is what the evaluations found, which the "findings" column summarizes. The evaluation gap concerns the first. Even then, to say that a policy "works" could mean three different things: that its effect is statistically significant, that it

[21] Eliminating the credits would reduce the EV share of new-vehicle sales in 2030 by about 6 percentage points (Buckberg and Cole, 2025), and repealing them together with the federal vehicle emissions standards would cut the battery-electric share from about 40 to 24 percent and battery-electric sales by about 40 percent (Jenkins, 2025). The model of Allcott et al. (2024) implies an own-price elasticity of demand for EVs of about -2.6.

is economically significant, or that it actually closes the welfare gap. The findings column speaks to the first two.

The categories with evidence that we would consider strong, based on credible quasi-experimental designs, are civilian R&D appropriations and the R&D tax credit. They amount to roughly 0.4 percent of GDP, about 30 percent of the comprehensive 1.31 percent portfolio. For both, the findings are favorable on the whole, though not uniformly so: within civilian R&D, measured returns range from 140–210 percent for nondefense appropriations to the near-zero spillovers of NASA spending (Section 2). A second group of instruments, roughly 0.91 percent of GDP, has been studied but with incomplete or conflicting findings: the semiconductor and advanced manufacturing credits (0.06 percent) are too recent for evaluation; defense R&D (0.48 percent) has been studied extensively, but credible designs reach conflicting conclusions, in part because they measure different outcomes in different settings (Section 2); and the targeted energy and production provisions (approximately 0.22 percent) rest on a sizable causal literature that concerns deployment, prices, demand, and early local employment effects but has not established aggregate or sectoral output or productivity effects (Shimberg, 2026). Deployment, prices, and demand are the outcomes the provisions were enacted to deliver; they are not the productivity and industrial-capacity outcomes that motivate the classification as industrial policy. Within the energy and production category, only the energy and vehicle credits (about 0.08 percent of GDP) have been studied; the FDII deduction and the outlays other than R&D (about 0.14 percent) have not been evaluated. The federal credit programs, approximately 0.15 percent of GDP, belong to this group too: their effects on the firms that borrow are well identified, but not their effects on what is produced, or by whom. On this analysis, we suggest that for only about one third of what can reasonably be labeled "industrial policy" is there strong evidence on effects, and even within the well-studied categories important questions like the aggregate R&D elasticity and the extent of relabeling are not resolved (Chen et al., 2021).[22]

The literature is not unanimous, even where we judge the evidence to be strong. Wallsten (2000), for instance, found that SBIR awards crowded out firms' own R&D spending dollar for

[22]The ratio also depends on how the credit programs are valued: 0.15 percent is the face value of new commitments, not their subsidy cost, which may be a small fraction of that, and part of the SBA lending is general small-business support with no industrial target (Erten et al., 2026). Excluding the credit programs and measuring the portfolio at fiscal cost (1.16 percent of GDP), the strong-evidence share rises to roughly 35 percent. Measuring defense R&D on OMB's narrower conduct-of-R&D basis (0.32 rather than 0.48 percent of GDP) gives about 35 percent with the credit programs and 40 percent without.

dollar, and Goolsbee (1998a) that a significant part of federal R&D spending goes into higher salaries for scientists and engineers. Hanson (2009) finds no effect of Empowerment Zones on employment once account is taken of how the zones were chosen, and Freedman, Kouchekinia, and Neumark (2025) find that the jobs created in Opportunity Zones were in large part offset by losses in nearby low-income neighborhoods. Benmelech and Monteiro (2023) find that much of the Export-Import Bank's aircraft financing went to airlines that could have borrowed privately. In weighing such results against those reported above, we have generally given more weight to the studies with sharper research designs.

Some of the instruments with the most developed evidence, general investment incentives and place-based programs, sit largely *outside* our headline industrial policy accounting at the federal level: the investment incentives we track separately as economy-wide capital subsidies, and the place-based programs are predominantly state and local. Including both would raise coverage to roughly half, or to between a third and two-fifths if the qualified business income deduction (§199A), which dominates the general-investment row and has little measured effect on investment or employment (Goodman et al., 2025), is not counted as strong evidence. Either way, a large share of industrial-policy spending flows without strong causal evidence on its efficacy, though for many of these tools there is a substantial theoretical basis.

The federal credit programs have been studied as small-business and export finance. SBA guarantees raise employment at the firms that receive them (Brown and Earle, 2017), and Export-Import Bank support raises exports, investment, and employment, most for firms facing financing frictions (Kabir et al., 2024), though some of it went to borrowers who did not need it (Benmelech and Monteiro, 2023). We grade this evidence as partial. It does not tell us whether federal credit has changed what is produced, or by whom, or what it has done for innovation and productivity. Furthermore, the Department of Energy's loan programs, which have an explicitly industrial aim, have not been evaluated at all.

The interaction effects across stacked instruments (R&D credits layered with energy credits, direct subsidies, and loan guarantees targeting the same firms and sectors) have received almost no empirical attention, despite being important features of the CHIPS Act and IRA.

3.2 Why the Gap Exists, and How Much It Matters

TABLE 1: INDUSTRIAL POLICY SPENDING AND THE CAUSAL-EVIDENCE BASE (FY2024)

Instrument category	% of GDP	Causal evidence	Findings	Key studies
Civilian R&D appropriations	0.32	Strong	High social returns to nondefense R&D; little spillover from NASA spending	Fieldhouse and Mertens (2024); Kantor and Whalley (2025)
R&D tax credit (§41)	0.08	Strong	Raises firm R&D, with an elasticity near one; part is relabeling or relocation	Rao (2016); Bloom, Van Reenen, and Williams (2019); Wilson (2009)
Semiconductor and advanced manufacturing credits (§48D, §45X)	0.06	Emerging	Early local employment gains in anticipation of the CHIPS Act as a whole; effects of the credits themselves not yet known	Erten, Stiglitz, and Verhoogen (2025); Miao (2024)
Targeted energy and production provisions	≈ 0.22	Partial (deployment)	Faster deployment, lower prices, responsive demand; aggregate productivity and capacity effects not established	Bistline, Mehrotra, and Wolfram (2023); Allcott et al. (2024); Gerarden, Reguant, and Xu (2026)
Defense R&D (RDT&E)	0.48	Contested	No aggregate productivity spillovers in US data; crowding-in of private R&D in OECD panels	Fieldhouse and Mertens (2024); Moretti, Steinwender, and Van Reenen (2025)
Federal credit programs (DoE LPO, EXIM, SBA, USDA)	≈ 0.15	Partial (borrowers)	Guarantees and export finance raise borrowers' employment, investment, and exports; no evidence on innovation, productivity, or the structure of production; DoE loans unevaluated	Brown and Earle (2017); Kabir et al. (2024)
Comprehensive portfolio	**1.31**			
Memo: outside the headline federal fiscal measure				
General investment incentives (§168(k), §179, §199A)	≈ 0.29	Strong (§168(k), §179)	Depreciation raises investment; employment effects mixed; §199A has little effect on either	Zwick and Mahon (2017); Ohrn (2019); Goodman et al. (2025)
Place-based programs (mainly state and local)	0.1–0.2	Strong	Local gains; aggregate effects uncertain; bidding for plants transfers the rents to firms	Busso, Gregory, and Kline (2013); Slattery (2025)

Notes: GDP shares are from the companion paper's FY2024 totals; rounded shares may not sum exactly. The targeted energy and production row is the difference between the broad and narrow measures: targeted tax provisions, among them the §48 energy investment credit and the FDII deduction (§250), and outlays other than R&D in the energy and the science and space budget functions. The cited evidence concerns only the energy and vehicle credits. Defense R&D is Department of Defense RDT&E outlays, and the credit programs are valued at the face value of commitments. "Strong" denotes multiple quasi-experimental studies with credible identification; "emerging" early quasi-experimental work on recent legislation; "partial (deployment)" credible quasi-experimental and structural evidence on deployment, prices, demand, and early employment, with little on productivity, spillovers, or industrial capacity; "contested" credible studies reaching conflicting conclusions; and "partial (borrowers)" credible evidence on the firms that receive credit, but none on innovation, productivity, or the structure of production. Classifications grade the state of the evidence (identification, agreement among credible studies, and coverage of the relevant outcomes), not whether the results are favorable, which the findings column summarizes; a well-identified null, such as Kantor and Whalley (2025), counts as strong evidence. The two rows below the line are memorandum items, excluded from the headline totals because the companion paper treats economy-wide investment incentives and state-local programs separately; most of the general-investment row is §199A. The place-based estimate is from Slattery and Zidar (2020) and Bartik (2019).

Why is there a gap, and can it be filled? Or are certain policies simply too difficult to study by causal methods? We see four reasons for the gap: the design of the instruments, the data and the time available, the limits of causal methods for aggregate questions, and the politics of evaluation. We take them in turn and then ask how much the gap matters. Some instruments are more amenable to causal microeconomic studies than others. Many features that make an instrument amenable to credible evaluation, such as eligibility thresholds, discretionary awards with close losers, and discrete legislative changes, are features that expose it to political scrutiny. The features that Surrey (1973) identified as letting tax expenditures escape budgetary oversight deprive researchers of identifying variation for the same reason: the instruments hardest to see in the budget tend to be the hardest to evaluate. With a program of discretionary grants like those of the CHIPS Act, one can compare those at the margin who got a grant with those who did not. A tax credit that every firm may claim on the same terms offers no such comparison, and in a democracy a tax law can hardly be written otherwise. The correspondence is not exact: the R&D tax credit is well studied because identification happens to be available, from state-level variation and from the interaction of federal rule changes with firms' histories (Rao, 2016), and defense R&D is highly visible but contested. But for the most part, instruments that lack thresholds, discrete reforms, or observable award decisions have gone largely unevaluated.

Part of the gap is simply a matter of data and of time. Tax returns are confidential, loan-level records of the credit programs are not assembled for research, and some of the instruments are too recent to have been evaluated. But evaluation catches up. In FY1985, instruments outside the narrow core totaled roughly 0.8 percent of GDP under the headline definitions with, at the time, no causal evaluation;[23] parts of that portfolio, notably federal R&D appropriations, have since been evaluated retrospectively (Fieldhouse and Mertens, 2024).

There are, moreover, questions that lie beyond the reach of these methods, whatever the instrument. Much of what industrial policy is meant to achieve, such as a change in the structure of production or greater resilience against supply disruptions, concerns aggregate outcomes that may take decades to appear. Designs that compare treated with untreated firms or places difference out economy-wide effects; and because support goes to sectors chosen for their prospects, the association between support and performance is uninformative (Rodrik, 2012; Juhász, Lane, and Rodrik, 2024). There is always, too, the question of the right counterfactual. Were there ways

[23] The figure is approximately 2.1 percent when the ITC and accelerated depreciation are included.

of spending the same money that would have had a larger effect, including a larger macroeconomic effect? For questions like these, theory, estimated structural models, and case studies have to carry more of the weight.

Finally, there is a political economy point. A rigorous evaluation can embarrass a program's sponsors (Pritchett, 2002), and non-evaluability may sometimes be less an accident than a feature of political design; the United Kingdom's National Audit Office found robust evaluation plans for only 8 percent of spending on the government's major projects (National Audit Office, 2021). But we think the issue is as much one of type I and type II errors, and of the worry that results will be misinterpreted. An evaluation might not be "right," at least for some conditions. A result that holds for one group of firms, or one period, or one outcome is easily read as a verdict on the program as a whole.

How much does the gap matter? To put matters in perspective, there is an absence of causal evidence in most other areas of expenditure, and in the absence of extensive pilot programs there will of necessity be a dearth of causal evidence on the efficacy of new programs. The deeper question is the marginal social value of more information. If we believe price theory, often we do not need such studies. The presumption is that changing a price induces the responses predicted by well-studied price elasticities. That is our Bayesian prior, and econometric studies are best framed as tests of it. We have a wealth of evidence that incentives work; is there any reason to believe that standard price elasticities do not apply to the particular incentives embedded in, say, the IRA? The evidence so far suggests not (Allcott et al., 2024). There may even be behavioral effects working in the same direction. If individuals want to contribute towards a better environment, and the subsidies encourage them to think about this, their response will be greater than standard price elasticities would predict. Additional evidence is most valuable where our priors are weakest, as with the credit programs, where the question is whether the government is financing what private lenders would not. Such a prior answers the first of our three questions, not the third. That firms respond to a price does not tell us whether the response was worth paying for.

How important should evaluability be in the selection of instruments? Should it lead us to have a bias towards instruments that can be tested? Such instruments entail either more discretion or arbitrary boundaries, both of which have costs of their own (Section 5). These are questions worth posing, even if we cannot answer them here.

The gap can be narrowed, then, but we doubt that it can be closed. Better access to data and

a requirement of evaluation would help with the credit programs and the discretionary awards, where evidence is most needed. For tax provisions open to all, and for the aggregate questions, policy will continue to have to rely on priors, theory, and indirect evidence.

The characterization of the literature in Table 1 may still evolve. Several recent policy changes, discussed in the Conclusion, remain to be examined.

4 Lessons from Other Countries

The instruments reviewed above have counterparts abroad, and some have been evaluated more cleanly there. We turn to that evidence with two purposes in mind. The first is to see what can be learned about instruments that have been little studied in the United States, credit above all. The second is to ask how far the US findings might carry over elsewhere. This is not a global survey of industrial policy, and our conclusions about other countries are correspondingly tentative. This section reviews evidence from other industrialized countries, then from developing countries; Section 5 draws the lessons.

4.1 Other Industrialized Countries

Some of the sharpest causal evidence on place-based support and investment incentives comes from program rules in other rich countries. Exploiting changes in the area eligibility rules of the United Kingdom's Regional Selective Assistance program (investment grants conditioned on job creation), Criscuolo et al. (2019) find that a 10-percentage-point increase in the maximum subsidy rate raises an area's manufacturing employment by roughly 10 percent, along with increases in investment and net entry, with the response confined to smaller firms (large firms took the subsidies without increasing activity) and no effect on total factor productivity. Exploiting a size-dependent change in investment tax credit rates in German manufacturing, Lerche (2025) finds that a 7.6 percent reduction in the user cost of capital raised treated firms' capital stock by 17.7 percent and employment by 12 percent, with positive spillovers generating roughly one additional manufacturing job, mainly at nearby firms and those linked through input-output networks, for every job created at treated firms.

Other evidence from industrialized countries appears in the instrument reviews above: the cross-country panels behind the R&D-credit elasticity (Bloom, Griffith, and Van Reenen, 2002;

Bloom and Griffith, 2001), the size thresholds in the United Kingdom's R&D relief exploited by Dechezleprêtre et al. (2023), and the OECD industry panel in which defense R&D crowds in private R&D (Moretti, Steinwender, and Van Reenen, 2025). The pattern resembles the US record: firm-level investment, employment, and R&D respond to incentives, while the aggregate questions (creation versus relocation, and whether spillovers justify the subsidy) remain open.

4.2 Developing Countries

Many industrial-policy instruments commonly used in developing economies (directed credit, targeted tax incentives, local-content requirements, special economic zones) resemble those the United States has long deployed. Writing before the recent revival of empirical research on industrial policy, Harrison and Rodríguez-Clare (2010) concluded that the evidence for "hard" interventions in developing countries was weak and largely observational.[24] Better identification has since shifted the picture. We treat the history of today's rich countries as part of this evidence, since their industrialization was itself a process of development. Juhász (2018) shows that temporary protection during the Napoleonic blockade induced persistent adoption of mechanized cotton spinning in France. Lane (2025) documents long-run gains in output and productivity from South Korea's heavy-industry drive, which persisted after the subsidies ended and extended to downstream users of the targeted intermediates. At the firm level, firms subsidized during the drive outgrew comparable firms for thirty years, and a quantitative model fitted to those estimates implies the policy raised aggregate welfare (Choi and Levchenko, 2025). These results are consistent with the older institutional literature on East Asian industrialization, in which policy loans at controlled rates and protection paired with export targets were the main instruments (Amsden, 1989; Wade, 1990; World Bank, 1993; Stiglitz, 1996; Stiglitz and Uy, 1996) and with the theoretical emphasis on learning externalities as the market failure industrial policy should target (Stiglitz and Greenwald, 2014).[25]

The developing-country literature also fills gaps that Table 1 records for the United States, credit above all. Results need not carry over between developing and advanced countries, which differ in many of the factors that make an industrial policy succeed or fail; this is the standard

[24] "Hard" interventions are tariffs, subsidies, and directed credit, as against "soft" facilitation such as information provision and coordination.

[25] There is also an enormous case-study literature, much of it very insightful. See, for instance, Cimoli, Dosi, and Stiglitz (2009), Stiglitz and Lin (2013), Stiglitz, Lin, and Patel (2013), and Noman and Stiglitz (2015).

external-validity problem of causal studies. Directed credit, studied in the United States only as small-business and export finance, is among the better-studied instruments in India: Banerjee and Duflo (2014) use a change in the size limit for firms eligible for India's mandated "priority-sector" bank lending to show that directed lending financed real expansions in output instead of substituting for other credit. Rotemberg (2019) studies subsidies to small firms in India and finds that expanding eligibility raised aggregate productivity by about 1 to 2 percent, even though in less-traded products the newly eligible firms' growth came almost entirely at the expense of ineligible competitors. On the question of where support does the most good, Liu (2019) shows that when sectors buy inputs from one another, market imperfections accumulate in upstream sectors, providing a formal rationale for the upstream targeting evident in East Asian practice.

Special economic zones now also have credible evaluations. Wang (2013) finds that Chinese special economic zones raised foreign investment and total factor productivity in treated municipalities, and Lu, Wang, and Zhu (2019), comparing villages inside and outside the zones, find that China's economic-zone program increased capital, employment, and output, in part through firm entry. Chaurey (2017) shows that location-based tax exemptions in India raised employment in treated regions. Aghion et al. (2015) find that Chinese subsidies raised productivity where they were allocated competitively and spread across firms within industries, and Barwick, Kalouptsidi, and Zahur (2025) find that much of China's support for shipbuilding was dissipated through excess entry and capacity. Evidence on procurement in developing countries is thinner; in Brazil, winning a government contract in a close auction causes lasting firm growth (Ferraz, Finan, and Szerman, 2015).

India's production-linked incentive (PLI) schemes, which pay firms a share of incremental sales of goods manufactured in India above a base year, resemble the §45X-style production credits, and like their US counterparts they remain largely unevaluated. Table 2 sets out, for each instrument, the evidence in each setting, the administrative capacity the instrument requires (for a tax credit, a corporate tax net that reaches the target firms), and its fiscal risk.

5 Relative Merits of Different Instruments

Industrial policy does most good where firms cannot appropriate the benefits of learning and of research and development, or where prices leave out social costs: carbon, in the case of green

TABLE 2: INDUSTRIAL POLICY INSTRUMENTS ACROSS CONTEXTS

Instrument	US evidence	Other-industrialized evidence	Developing-country evidence	Administrative prerequisites	Fiscal-risk profile
Public R&D and innovation grants	High social returns to nondefense R&D; grants relax financing constraints (Fieldhouse and Mertens, 2024; Howell, 2017)	Defense R&D can crowd in private R&D across OECD countries (Moretti, Steinwender, and Van Reenen, 2025)	Large yield and income gains from Green Revolution agricultural research (Gollin, Hansen, and Wingender, 2021); little evidence on industrial R&D programs	Research institutions; competitive grant administration	Capped; on-budget
R&D and production tax credits	Long-run R&D elasticity near one (Bloom, Van Reenen, and Williams, 2019); §45X-style production credits too recent to evaluate	R&D responds at UK size thresholds (Dechezleprêtre et al., 2023); cross-border reallocation (Bloom and Griffith, 2001)	Relabeling at eligibility notches in China (Chen et al., 2021); India's PLI production credits not yet evaluated	Broad formal tax base; audit capacity	Open-ended if uncapped
Investment incentives	Large investment responses (Zwick and Mahon, 2017; Ohrn, 2019)	Capital and employment gains with local spillovers in Germany (Lerche, 2025)	Aggregate productivity gains, despite crowd-out of competitors, from size-based small-firm subsidies in India (Rotemberg, 2019)	Corporate tax net covering target firms	Open-ended if uncapped
Directed credit and guarantees	Raise borrowers' employment and exports; effects on the structure of production not studied (Table 1)	—	Directed lending relaxed real credit constraints in India (Banerjee and Duflo, 2014)	State banking capacity; supervision	Contingent liabilities
Sectoral targeting and place-based support	Local gains; bidding for firms transfers the rents to them (Busso, Gregory, and Kline, 2013; Slattery, 2025)	Employment gains from changes in subsidy eligibility in the UK (Criscuolo et al., 2019)	Persistent firm and sectoral gains from Korea's heavy-industry drive (Lane, 2025; Choi and Levchenko, 2025); upstream-targeting rationale (Liu, 2019)	Discretion paired with accountability; performance standards	Capped but capture-prone

Notes: Fiscal-risk profiles refer to the delivery mechanism: tax-based instruments that operate automatically are open-ended entitlements whose ultimate cost depends on take-up (Bistline, Mehrotra, and Wolfram, 2023), while grant and credit instruments are capped in appropriation but carry discretion (grants) or contingent liabilities (guarantees). PLI denotes India's production-linked incentive schemes. Administrative prerequisites summarize the state capacity each delivery channel presupposes.

industrial policy, and resilience and national security, in the case of the CHIPS and Science Act. How much of that good a policy delivers depends in part on the instrument. We compare the instruments on selection, fiscal risk, administrative cost, conditionality, political durability, and transparency, and then turn to what the US record implies for other countries.

Critical to the success of any industrial policy is *selection*: whether support reaches the activities that generate the most learning and the largest spillovers. Instruments select in different ways. One advantage of tax credits is that they harness a self-selection mechanism: only firms that undertake the subsidized activity claim them. But a credit cannot tell an activity with large spillovers from one without. Discretionary grants can target more finely, but only as well as the agency's information allows, and they invite lobbying (discussed below).

The alternative instruments also carry different fiscal risks (Table 2). Tax credits that operate automatically are typically uncapped; for a government facing hard borrowing constraints an uncapped credit that succeeds in generating take-up becomes a fiscal problem. Direct grants are capped, but their allocation is discretionary and exposed to political capture. Trade and credit interventions pose less of a financial burden; indeed tariffs may actually generate revenues, though they raise prices for consumers and for the firms that use the protected goods, and, like credit, they are open to political capture. But poorly designed credit programs can also be costly because of non-repayment. Partial government guarantees enlist private incentives in screening borrowers, for repayment if not for spillovers, and government equity investments and warrants accompanying debt issuance enable government to capture more of the upside of any government investment.

An important feature is the cost of administration and compliance. Tax-based instruments that operate automatically are administratively simple; discretionary grant programs permit targeting but depend on continuing appropriations and often contentious selection processes, with, as we have noted, opportunities for politics to weigh in (discussed further below). But both kinds of programs need oversight: abuses abound.

Well-targeted programs, whether through tax expenditures or grants, have to have "conditionality." Discretion is also what makes conditionality possible, but it increases administrative costs. The East Asian miracle countries allocated support through contests, paired it with performance standards (export targets, capacity and quality milestones), and withdrew it from firms that missed them (Amsden, 1989; Wade, 1990; World Bank, 1993), militating against subsidy cap-

ture. The current US programs carry weaker versions: CHIPS awards disburse against negotiated milestones, and the IRA credits pay only on production or investment actually undertaken; production credits, which pay for output rather than capacity, build in a condition that investment credits lack (Aldy, Gerarden, and Sweeney, 2023). Sunset clauses that end support automatically can also provide discipline (Rodrik, 2004; Juhász, Lane, and Rodrik, 2024).

Political economy considerations are central. All forms of industrial policy pose governance risks, some more than others, and some countries are better able to manage these risks. The lack of transparency of tax expenditures is, for instance, problematic, and countries with smaller formal sectors and weaker oversight may have greater problems in design and impartial implementation. The durability of policy is itself an element of policy design. Political durability may track the breadth of an instrument's constituency rather than the importance of the goal or the effectiveness of the mechanism. While Biden had thought that he had "Trump-proofed" the IRA EV tax credits, with many of the benefits going to "red" states, ideology trumped self-interest. The projected declines in electric-vehicle sales and investment following the repeal of federal EV policies (Jenkins, 2025; Buckberg and Cole, 2025) illustrate how sensitive long-horizon capital formation is to the expected persistence of incentives. That consideration bears on the CHIPS and IRA programs, whose returns depend on investments that pay off over decades.

Of course, just as there are market failures, so too there are government failures (Krueger, 1990): governments may lack the information to direct investment well (Hepburn, 2010), support creates rents that firms compete for (Krueger, 1974; Helm, 2010), and agencies may be unable to withdraw support from projects that have failed. The results we have reported are, though, "net" results. They are the effects of programs directed at a market failure but shaped by our imperfect politics. The programs might have been even better designed, and still more effective, in the absence of such politics. The clearest examples above, the bidding for plants (Slattery, 2025) and the subsidies that drew inefficient entrants into Chinese shipbuilding (Barwick, Kalouptsidi, and Zahur, 2025), argue for better design rather than against industrial policy *per se*: for rules rather than discretion where administrative capacity is weak, for competition among recipients (Aghion et al., 2015), for transparency about who receives what, and for support that expires unless it is renewed.

Transparency helps to ensure good outcomes, especially in the many countries where it is hard to create the kinds of independent institutions that can administer industrial policies with-

out political interference. Much industrial policy, in developed and developing countries alike, is delivered through channels that are hard to observe: tax expenditures in the United States, subsidized credit and below-market inputs elsewhere. This makes undertaking good causal analyses all the more difficult. Kalouptsidi (2018) has to infer unobserved Chinese shipbuilding subsidies from production and cost data; assessing effectiveness requires first measuring the policy.

What does the US record imply for other countries? Presumably the responses of firms to incentives would be similar in other advanced countries; one would need to think of a reason why that might not be true, and the British and German studies discussed in Section 4 suggest that there is none. Whether the aggregate gains would be similar depends on institutions and on the structure of production, about which the evidence says less. For developing countries, objectives and context are likely to differ, and so too will the appropriate choice of instruments. Still, there are some general insights. One concerns incentives. Firms have responded to changes in the cost of research, capital, and credit wherever the response has been measured. A second concerns resources. Instruments differ markedly in their fiscal cost and in the administrative capacity they presuppose (Table 2). A third concerns information. Governments need to know what they are spending and what it achieves, and this is hardest for the least visible instruments.

The instrument mix that is optimal in one context will not be in another. The US delivers much of its support through tax expenditures, alongside a small layer of federal credit guarantees: instruments that presuppose a broad corporate tax base, sophisticated tax administration, and deep credit markets.[26] Direct grants, trade policy, and directed credit have historically played the corresponding role in developing economies.

Fiscal instruments also increasingly interact with trade. The IRA's local-content requirements condition tax credits on domestic or friendly-country production, hoping to reshape global value chains in solar, batteries, and electric vehicles (Juhász, Lane, and Rodrik, 2024), much as China's earlier solar subsidies reorganized the global solar industry (Bañares-Sánchez et al., 2026). For developing countries integrated into those value chains, the question is how to position themselves within supply chains being reorganized by the subsidies of others. The post-2018 subsidy escalation has been driven mainly by advanced economies (Evenett and Fritz, 2022; Juhász et al., 2025), which can finance support at a scale and a cost of borrowing that developing countries

[26] Even in deep markets credit is rationed, especially for small and innovative firms (Stiglitz and Weiss, 1981). The Department of Energy's $465 million loan to Tesla in 2010, repaid in 2013, is the best-known US example; whether it financed what private lenders would not is exactly the question that has not been studied systematically.

cannot. For most developing countries, the practical question is where in the reorganizing value chains modest, well-targeted support can still shift comparative advantage, and what they can do within the international rules of the game, rules which appear increasingly irrelevant, however, for the major powers.

Another lesson is that countries now constructing industrial strategies for the clean-energy transition should build evaluation into policy design from the outset. There are well-known ways of doing so: published eligibility thresholds, scored applications where grants are oversubscribed, phased introduction, sunset provisions paired with a required evaluation, and access for researchers to tax and loan records.[27]

6 Conclusion

While industrial policy is back, and the presumption that one can simply rely on markets is decidedly out of favor, it is in flux. The Trump administration eviscerated key parts of the IRA, and has remade the CHIPS Act, converting grants already awarded to Intel into a 9.9 percent equity stake (Intel Corporation, 2025) and renegotiating other awards. The One Big Beautiful Bill Act (OBBBA) of 2025 raised the semiconductor investment credit (§48D) from 25 to 35 percent, but terminated the §30D electric-vehicle credit as of September 2025, and curtailed the clean-electricity and clean-hydrogen credits and the credit for wind components.[28] The concurrent pause in new Loan Programs Office originations also affects the credit-program category. At the same time, Trump has put trade policy at the center of industrial policies aiming to bring manufacturing back to the United States. While the literature on causal analyses of the recent tariffs is at an early stage, the question of their effects on manufacturing is more complicated than a look at the aggregate statistics might suggest. Manufacturing employment has increased little, and the recovery in output since late 2025 is concentrated in the high-technology industries sup-

[27] Evaluations are more likely when an authority outside the spending ministry, typically the finance ministry, makes an evaluation plan a condition of funding. This was the purpose of the Evaluation Task Force that the United Kingdom's Treasury and Cabinet Office set up in 2021.

[28] §30D provided a consumer credit of up to $7,500 for new electric vehicles. Wind and solar projects lose §45Y and §48E, the IRA's technology-neutral clean-electricity production and investment credits, unless they began construction by July 4, 2026, or enter service by the end of 2027; in June 2026 a federal court vacated the IRS notice that had tightened the begin-construction test (*Oregon Environmental Council v. Internal Revenue Service*, D.D.C., June 6, 2026). The clean-hydrogen credit (§45V) closes to projects beginning construction after 2027, and the per-unit credit for domestically manufactured clean-energy components (§45X) ends for wind components sold after 2027.

plying the build-out of artificial-intelligence capacity.[29] But what is the right counterfactual? And how should we think about the investment pledged in response to the threat of still higher tariffs? Is it really new investment, and was it the tariffs that motivated it, or other factors? It is too soon to say. Future administrations are likely to attempt to revive industrial policies in a more systematic way, both in the US and elsewhere, and there will be an attempt to learn from the past.

This review has shown that well-designed industrial policies can affect patterns of production and levels of innovation. Their effectiveness, however, depends on design. There remain large gaps in our knowledge. We are not making a judgment about the overall success of US industrial policy. But the studies that have attempted to assess the effectiveness of particular instruments by and large show positive effects, and, so far as the response of firms to incentives is concerned, there is nothing special about the circumstances of these studies that would lead one to expect the results not to hold more broadly.

The most credible and most consistent evidence in this review concerns the public financing of research: nondefense federal R&D generates gross social returns estimated at 140–210 percent (Fieldhouse and Mertens, 2024), R&D grants to small firms raise patenting and relax financing constraints (Howell, 2017), and R&D tax credits increase research spending with a firm-level long-run elasticity of roughly one, with real downstream effects on entrepreneurship and employment (Bloom, Van Reenen, and Williams, 2019; Fazio, Guzman, and Stern, 2020; Lucking, 2019). Investment tax credits and accelerated depreciation reliably stimulate capital formation (Zwick and Mahon, 2017; Curtis et al., 2022; Lerche, 2025). Other interventions appear less effective: defense R&D and NASA spending show little measurable productivity spillover in US aggregate data within a fifteen-year horizon (Fieldhouse and Mertens, 2024; Kantor and Whalley, 2025); state-level tax credits partly reallocate activity across jurisdictions rather than create it (Wilson, 2009); some measured responses reflect relabeling of existing expenditures or inframarginal investment (Chen et al., 2021; Corinth and Feldman, 2024); and when governments compete for mobile plants, the firms capture the gains (Slattery, 2025). We would argue that these smaller or mixed effects tell us more about design and measurement than about the case for intervention.

[29] Manufacturing output rose 0.9 percent in the twelve months to August 2026: 12.5 percent in the high-technology industries (computers, communications equipment, and semiconductors) and 0.5 percent in the rest (Federal Reserve, Industrial Production and Capacity Utilization, G.17, September 18, 2026). The Federal Reserve attributes the strength in computer and electronic products to AI-related investment (Monetary Policy Report, July 2026). Manufacturing employment in September 2026 was 72,000 above its low of December 2025 (Bureau of Labor Statistics, The Employment Situation, October 2, 2026). The tariffs of 2018–19, which have been studied, lowered manufacturing employment on net: the protection of domestic producers was more than offset by higher input costs and retaliation (Flaaen and Pierce, 2024).

The record suggests additional, broader lessons. The returns to public support depend on the spillover potential of the activity being subsidized, and basic and general-purpose research generates large external benefits. Aggregate spending totals, including those we construct in the companion paper, are therefore only a starting point for welfare analysis. The distinction between creating and reallocating economic activity is central to interpretation, especially in place-based policies. Many of the cleanest estimates identify local or compositional effects; whether these translate into aggregate gains depends on spillovers, migration responses, and general-equilibrium adjustments that most credible research designs cannot capture. The contrast between US place-based programs, whose well-identified gains are local and whose aggregate effects remain uncertain, and the persistent structural transformation documented for South Korea's heavy-industry drive (Lane, 2025) suggests that much depends on whether targeted sectors feature strong learning externalities and input-output linkages (Stiglitz and Greenwald, 2014).[30]

The evaluation gap quantified in Section 3 suggests an agenda for future research. We would highlight four areas. The first is lending. Causal studies of its effects on innovation, productivity, and the structure of production are nearly absent, and its true costs are especially hard to estimate; loan-level records would allow approved applicants to be compared with those narrowly turned down. The second is the production and investment credits of 2022, where the staggered phase-outs legislated in 2025 have created differences in exposure that did not exist while the credits were open to all. The third is the interaction of instruments stacked on the same firms. And the fourth is the set of aggregate questions that firm- and place-level designs cannot reach. More broadly, we need to know much more about the relative effectiveness of different interventions, their costs, the circumstances that determine costs and benefits, and how interventions interact with each other and with other government programs and policies.

[30] A related, contested literature measures the complexity and interlinkage of what economies produce (Hidalgo and Hausmann, 2009; Tacchella et al., 2012).

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